

Guideline Title: Crush Injury and Crush Syndrome

Guideline Number: GUID-3203-T003

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 07/01/2026	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** A crush injury is a direct injury resulting from pressure. Crush syndrome is the manifestation of rhabdomyolysis caused by prolonged continuous pressure on muscle tissue. The goal is to identify crush syndrome and begin therapy early to prevent cardiac decompensation by early recognition and management of hyperkalemia.
- II. **DEFINITIONS:**
- III. **DEPARTMENT GUIDELINE:**
 - A. Perform a general trauma assessment. Evaluate and support airway, breathing, and ventilation concerns.
 - B. Assess for absent pulse or weak rapid pulse, capillary refill, pale, clammy, or cool skin, unusual pain, and signs of shock.
 - C. Initiate and ensure adequate IV/IO access.
 - D. Review or consider obtaining a 12-lead ECG.
 - E. Keep patient warm.
 - F. Consider infusing 1 L (adult) or 20 ml/kg (pediatric) warm 0.9% NS or Lactated Ringers wide open prior to or shortly after extrication. Use caution in setting of congestive heart failure and end-stage renal disease.
 - G. If Creatinine Kinase levels greater than 6000 IU/L or a large mass of skeletal muscle (one or more limbs, with or without trunk or pelvis) has been crushed for ≥ 4 hours, consider Sodium Bicarbonate
 - i. If prior to extrication
 1. Adult: **Sodium Bicarbonate 8.4% (50 mEq) IV/IO**
 2. Pediatric: **Sodium Bicarbonate 8.4% 1 mEq/kg IV/IO**, max dose 50 mEq
 3. Infant: **Sodium Bicarbonate 4.2% 1 mEq/kg IV/IO**
 - ii. If following extrication
 1. Adult: **Sodium Bicarbonate 8.4% (40 mEq/250 ml D5W) infusion** at 250ml/h.
 - a. Remove 40 ml of fluid from bag prior to adding bicarb

Guideline Title: Crush Injury and Crush Syndrome

Guideline Number: GUID-3203-T003

2. Pediatric: **Sodium Bicarbonate 8.4% (40 mEq/250 ml D5W) infusion at maintenance fluid rate** per GUID3203-PED001 – *General Management of the Pediatric Patient*

- a. Remove 40 ml of fluid from bag prior to adding bicarb

- H. Immobilize the limb. Do not elevate.
- I. If signs of inadequate perfusion, treat per GUID3203-M011 - *Hypotension*.
- J. If hyperkalemia is known or suspected (bradycardia, QRS > 0.12 ms, or peaked T-waves), treat per GUID3203-M006 - *Electrolyte Disorders – Potassium Imbalance*.
- K. Treat pain per GUID3203-G005 - *Analgesia and Sedation Management* and anxiety per GUID3203-G006 *Anxiety- Combative Patient Management*.
- L. If after discussion with those providers leading the extrication attempt it is determined that the time required for extrication will be a risk to the patient's life, the flight team should consider contacting medical control regarding bringing an appropriately trained physician to the scene to consider amputation of an entrapped limb.

IV. DOCUMENTATION:

- A. EMS Charts

V. REFERENCES: